Using LLMs for post-OCR correction on historical French texts: A case study using synthetic data

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Introduction

Automated text recognition systems have recently achieved remarkable accuracy, with character error rates (CER) as low as 0.01% (Reul et al., 2018). However, transcription quality remains a persistent challenge for degraded or complex layouts, particularly in historical documents. These often feature ornate fonts, physical degradation, and unconventional page layouts, making OCR particularly error-prone.

As part of the Valentin Haüy Digital Library project[1], our goal is to establish a pipeline for the digitization and OCR of texts, as well as their correction and enriched digital editing. In this paper, we focus on the issue of post-OCR correction.

State of the art

Three main approaches to post-OCR correction are prominent: lexical methods, machine learning techniques, and language models (Petkovic et al., 2022). Lexical approaches are declining, but machine learning and pre-trained models show promise.

Pre-trained LLMs have garnered interest. Thomas et al. (2024) showed that fine-tuned Llama-2 models reduced CER by 50% in the British newspaper dataset BLN600. Boros et al. (2024) reported mixed results with pre-trained models evaluated with the Post-Correction Improvement Score (PCIS). They noted quality degradation across models and prompts but slight improvements with sentence-level tokenization and zero-shot prompts with explicit instructions for the GPT-4 model.

Jonathan Bourne (2024) emphasized diverse training data for OCR correction. Analyzing OCR error distributions with a character-level Markov corruption process, he developed the 'ScrambledText' Python library to simulate OCR errors. Bourne found that augmenting datasets with synthetic errors improved performance on real-world data.

Our study builds on these insights to explore the application of advanced LLMs for post-OCR correction, focusing on the unique challenges presented by French historical texts.

OCR evaluation metrics vary, sometimes incorporating semantic or morphosyntactic features based on evaluation needs (Roux et al., 2022). This study used WER, CER, and PCIS (Boros et al., 2024). PCIS (-1 to 1) reflects improvement (positive) or degradation (negative).

Dataset & Preprocessing

The dataset [2] used is from the ICDAR 2017 competition corpus (Chiron et al., 2017a; 2017b), containing historical OCR transcriptions aligned with ground truth (GT). Texts were filtered by publication year (19th century), type (monograph), and language (French).

After character-level preprocessing[3], OCR and GT texts were tokenized with pySBD (Sadvilkar & Neumann, NLPOSS 2020), ensuring identical phrase splits. A similarity threshold of 85% was applied via the 'FuzzyWuzzy' module to verify sentence pairing.

Table 1 summarizes the dataset, detailing sizes of training, development, and test sets with CER and WER medians. A filter limited zero CER cases to 15%, ensuring models learned to preserve accurate OCR outputs. Consequently, 89% of cases showed 0-0.03 CER between OCR and GT pairs.

	Train	Dev	Test
Phrases	2.63k	336	301
Characters	434k	48k	46k
CER	0.0126	0.0147	0.0139
WER	0.0552	0.0632	0.0621

Table 1: Summary of the dataset's characteristics

To increase CER diversity, the 'ScrambledText' Python module introduced realistic OCR errors based on observed error probabilities. This crea-

ted a 'Sentence_OCR_corrupted' version, raising CER in 51% of sentences to 0.07 – 0.11.

Case Study

We used the Llama-3.2-3B-Instruct model for OCR correction with a resource-constrained setup using Low-Rank Adaptation (LoRA) for efficient fine-tuning. Experiments were conducted on two instruction datasets: "natural" with authentic OCR errors and "synthetic" augmented with simulated errors from the 'ScrambledText' library.

Prompt template
You are performing a task to correct OCR-processed text. 1. Correct OCR errors (missing letters, accents) in the text provided from the Gallica digital library. 2. Preserve the 19th-century style and language, avoiding modernization. 3. Do not alter line break markers (e.g., seu-lement, oc-tobre). 4. Return the corrected text only between the tags `<RESULT></RESULT>`.

Table 2 : Prompt translation from French

This approach evaluated the impact of different training data on model performance. Task-specific details were included in prompts to improve contextual understanding, and formatting constraints required wrapping corrected text in XML-style tags.

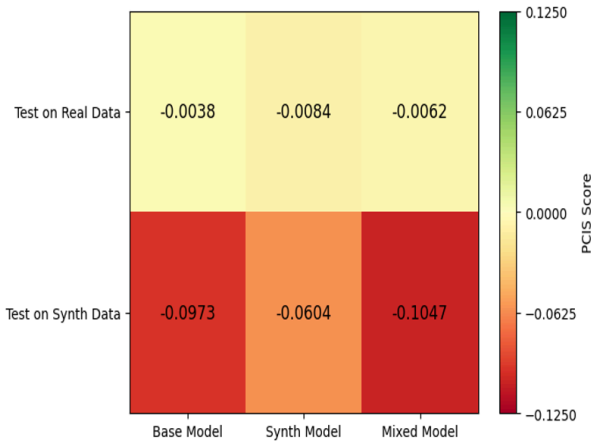
Using LoRA, we adapted the model efficiently without updating all parameters, reducing computational demands. Fine-tuning employed 4-bit quantization for memory efficiency with Hugging Face TRL's SFTTrainer, evaluating every 30 steps and applying early stopping after 7 steps without improvement.

3 models were prepared for this study:

- **Base model** trained on "natural" errors of the original dataset;
- **Synth model** trained on phrases with "synthetic" errors;
- **Mixed model** trained on 80% "natural" and 20% "synthetic" errors.

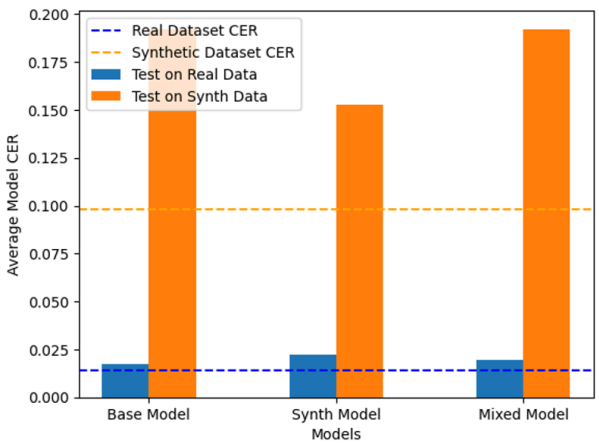
Results

All three final models showed negative PCIS scores, suggesting that the fine-tuning process did not improve the OCR output:

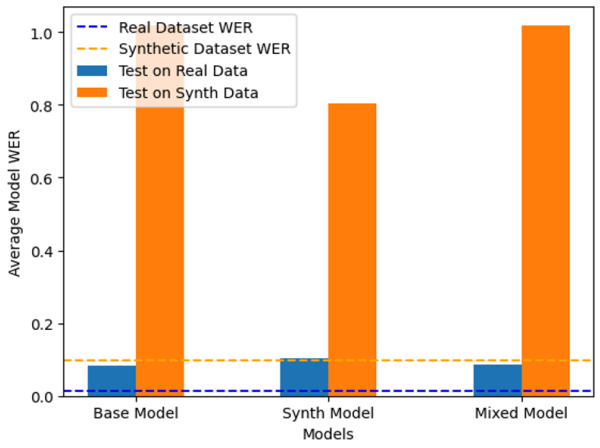


Average of post-correction improvement score (PCIS) per model and dataset evaluation type

Figure 2 & 3 show that the best-performing model, which was fine-tuned on and tested with real data, did not achieve significant improvements in terms of CER and WER, indicating that the LLMs were unsuccessful in reducing these error rates for this dataset.



Comparison of Average Model Character Error Rates



Comparison of Average Word Error Rates

These results highlight the inherent challenges of generalization that arise when using LLMs to correct OCR output on historical documents. Anecdotally, LLMs – especially when used through the chatbot interfaces they power – are capable of OCR text correction at a fine-grained level, but even then can be prone to overcorrection and hallucinations. At the scale of large corpora, LLMs still fall short of surpassing the state-of-the-art in post-OCR correction, particularly for historical texts.

Perspectives & limitations

This study illustrates the difficulties of achieving positive results in real-world conditions, where resource constraints often prevent the full potential of advanced techniques from being realized. Although this result does not have an immediate impact on our project (since the expected digital edition at the end of the project will still require manual correction after the automatic process), we aim to focus in the near future on optimizing training strategies and exploring alternative approaches. This includes developing a custom dataset, integrating more sophisticated error simulations, or leveraging greater computational resources.

Data availability

The code and data resources used in this work are available at : https://github.com/obtic-sorbonne/ocr_correction .

Notes

1.
2. . See also and .
3. Symbols such as "@" mark alignment points, while "#" represents uncertainties in the dataset. Since no images of the original documents were included, we first preprocessed the GT by replacing "#" symbols with the corresponding OCR characters to maintain input coherence. This ensures that LLMs can identify and learn patterns in corrupted text, leaving untouched the corrupted text difficult to reconstruct.

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